Rana I. Massoud
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RESEARCH PROFILE

PhD candidate in Computational Physical Chemistry with multidisciplinary research experience spanning **environmental toxicology, endocrine disruption, analytical chemistry, bio-organic chemistry, molecular modeling, and experimental biological evaluation**. Over 6 years of research experience investigating chemical-biological interactions and mechanisms of toxicity using computational and experimental approaches. Doctoral research focuses on **endocrine-disrupting chemicals (EDCs), PFAS, nuclear receptor interactions, molecular mechanisms of environmental chemical toxicity, and computational approaches for complex chemical mixtures**. Previous experimental research includes **GC-MS, NMR, FTIR, chemical characterization of natural products, and cell-based biological assays**. Experienced in integrating chemical, biological, and computational data to characterize bioactivity and support environmental health and risk assessment. Seeking to apply this interdisciplinary background to **translational toxicology, endocrine-disrupting chemical exposome research, mixture toxicity, analytical toxicology, and molecular mechanisms of human health effects**.

EDUCATION

- **PhD in Computational Physical Chemistry** at **Michigan State University**
  GPA: 3.63/4.00 | Expected May 2027 | East Lansing, MI, USA
- **MS in Bio-Organic Chemistry** at **Beirut Arab University**
  GPA: 3.72/4.00 | July 2022 | Beirut, Lebanon
- **BS in Chemistry** at **Beirut Arab University**
  GPA: 3.82/4.00 | January 2017 | Beirut, Lebanon

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Physical Chemistry at **Michigan State University**
Advisor: Dr. Angela K. Wilson | August 2022–Present | East Lansing, MI, USA

Environmental Toxicology, Endocrine Disruption, and Molecular Mechanisms

- Investigated the interactions of **16 per- and polyfluoroalkyl substances (PFAS)** with estrogen receptor α and β in rainbow trout (*Oncorhynchus mykiss*) using *in silico* modeling methods to provide mechanistic insight into endocrine disruption in aquatic organisms.
  - This work represented the **first comprehensive computational investigation** of a diverse panel of PFAS interacting with both estrogen receptor isoforms in this model organism.
  - The findings established the **collaborative studies** investigating the **biotransformation products** of PFAS Diols, integrating molecular modeling with metabolomics analyses to evaluate the estrogenic potential of **15 PFAS biotransformation products**.
  - This work expanded my research interests toward integrating molecular modeling with systems-level approaches for understanding environmental chemical toxicity.
- Characterized the **steroid, nonsteroid and lipid mimicry** of aqueous film-forming foam (**AFFF)-associated PFAS** to *Daphnia magna* nuclear receptors (EcR and HR96) through in silico modeling
  - Mechanistic understanding of PFAS toxicity in an ecologically important **bioindicator species** and contributed computational evidence supporting **new approach methodologies** (NAMs) for environmental hazard and **ecological risk assessment**.
  - Initiated **interdisciplinary collaborations** investigating short and ultrashort chain PFAS **mimicry** of **fatty acids** through the integration of molecular modeling and metabolomics.

- Investigated the **estrogen** and **androgen mimicry** of 8 PFAS and PFAS mixtures in fathead minnow
    - Molecular insights into the binding mode of **eight** monomeric PFAS (legacy and emerging) to estrogen (ER) and androgen receptor (AR).
    - Enhanced sampling of the **binding pathway** of monomeric and **binary mixtures** of PFAS to the AR to investigate the effects of chain length and headgroup on the toxicity mechanisms of PFAS binding (additive, competitive or synergistic) and the binding of **endogenous ligands.**
- Studied the **estrogenic molecular initiating events of food and textile dyes** in humans
    - Prediction of the binding potential and structural rearrangements of the human estrogen receptors upon binding azo and anthraquinone dyes compared to estradiol (E2)

Computer-Aided Drug Discovery and Molecular Toxicology

- Designed and evaluated novel inhibitors targeting **multiple** family proteins using an integrated computational drug discovery pipeline combining molecular docking, molecular dynamics simulations, binding free-energy calculations, residue interaction analyses, fragment- and structure-based drug design.
- Prioritized lead anti-tuberculosis compounds for experimental validation by identifying candidates with favorable binding stability and interaction profiles, reducing the number of compounds requiring experimental screening and accelerating lead optimization.
- Investigated the structural basis of ligand recognition and binding mechanisms across multiple therapeutic targets to support the rational design of compounds with improved affinity, stability, and selectivity.
- Applied quantitative structure–activity relationship (QSAR) modeling to identify molecular descriptors associated with DdN inhibitor activity and guide the design of new analogues with enhanced predicted potency.
- Investigated the anticancer potential of novel GPR68 (OGR1) inhibitors by characterizing ligand–receptor pharmacophore interactions and evaluating their structural stability through molecular dynamics simulations and binding free-energy analyses.

Method Development and Scientific Computing

- Designed and optimized computational workflows for molecular dynamics simulations using AMBER and developed automated analysis pipelines for enhancing the efficiency of structural and binding analyses (RMSD, RMSF, hydrogen-bond occupancy, binding energies, residue energy decomposition, principal component analysis, and trajectory processing)
- Performed large-scale molecular simulations on high-performance computing (HPC) systems, integrating bash scripting and automation
- Collaborated with multidisciplinary research teams in computational chemistry, environmental toxicology, and drug discovery, contributing to project design, data interpretation, scientific publications, conference presentations, and grant preparation.

Graduate Research Assistant - BioOrganic Chemistry at **Beirut Arab University**
Advisor: Dr. Mohammad H. El-Dakdouki | 2019–2022 | Beirut, Lebanon

- Characterized the chemical composition of *Lavandula angustifolia* and *Lavandula* × *intermedia* essential oils cultivate in Lebanon using FTIR, NMR, and GC–MS to identify bioactive constituents and lead evidence-based industrial manufacturing.
- Evaluated antibacterial, antioxidant, anti-inflammatory, antiproliferative, and hemolytic activities using complementary biochemical and cell-based assays.
- Correlated chemical composition with biological activity to identify compounds contributing to therapeutic potential.

- Established a strong interdisciplinary foundation in analytical chemistry, natural products, and experimental biological evaluation that now complements my computational research in toxicology and molecular modeling.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant at **Michigan State University**
Courses: Physical Chemistry I | January 2023 – May 2023 | East Lansing, MI, USA
General Chemistry Laboratory | August 2022 – December 2022 | East Lansing, MI, USA

- Supported student understanding of course material and assisted with homework, exam preparation, and grading of undergraduate Physical Chemistry I, by holding weekly sessions of 60 students covering topics such as thermodynamics and kinetics
- Led weekly General Chemistry laboratory sessions for 40 undergraduate students, providing instructions in experimental design, chemical safety, and laboratory techniques, guiding data collection and analysis, enforcing safety protocols, and grading lab reports and quizzes while providing feedback to improve scientific writing and presentation skills.
- Cultivated a supportive learning environment by answering questions and offering one-on-one assistance through weekly office hours
- Strengthened communication and mentoring skills through direct academic support to students from diverse cultural and academic backgrounds

Flight Attendant at **Middle East Airlines (MEA)**
MEA Headquarters | September 2017 – May 2022 | Beirut, Lebanon

- Served as a flight attendant for the national carrier of Lebanon and member of SkyTeam and IATA, MEA, ensured passenger safety and comfort by performing pre-flight inspections and conducting safety demonstrations, handling emergencies, and maintaining high service standards
- Collaborated with multicultural crew teams, enhancing communication, teamwork, and adaptability
- Adapted to irregular schedules, new destinations, and diverse cultural environments

Chemistry Teacher at **English Academy**
August 2016 - June 2017 | Mount Lebanon, Lebanon

- Taught high school chemistry as a part-time instructor, delivering lessons aligned with national curriculum standards, developed lesson plans and lab exercises to explain core concepts such as chemical reactions, stoichiometry, and thermodynamics, prepared and graded assessments
- Cultivated strong classroom management and communication skills

Intern at **The Ministry of Environment, Chemical Safety Department**
Supervisor: Lama Mghames | May-August 2016 and May-August 2015 | Beirut, Lebanon

- Assisted in a data-driven project to evaluate and propose upgrades for aging electricity columns, focusing on implementing newer, environmentally friendly materials and designs to reduce ecological impact and improve sustainability
- Conducted environmental toxicity assessments of persistent organic pollutants (POPs) following the open waste incineration crisis in Lebanon; analyzed data collected from soil, water, and air samples to evaluate contamination levels and potential health risk in different regions of Lebanon
- Aligned findings with international policy frameworks, including the World Bank guidelines and the Stockholm Convention.

SKILLS and TECHNIQUES

- **Modeling:** Molecular docking, pharmacophore modeling, virtual screening,

molecular dynamics simulations (classical MD, steered MD, umbrella sampling), free energy calculations (MM-PBSA, MM-GBSA, and FEP methods)

- **Software:** AMBER, MOE, Chimera, VMD, AutoDock Vina, Schrodinger, Gaussian, AlphaFold, I-TASSER, SwissModel, NERDD, Swiss ADME
- **Coding & Data:** Python, RDKit, Matplotlib, Seaborn, Scikit-learn
- **Experimental:** NMR, FTIR, GC-MS, MS, Cell viability (MTT), antimicrobial & antioxidant assays

PRESENTATIONS and WORKSHOPS

- **"In silico Investigation of PFAS Effects on Reproduction and Growth in Freshwater Organisms"** Biophysical Society Symposium (BPS) | October 2025 | San Francisco, CA, USA
- "***In silico* Investigation of PFAS Accumulation and Toxicities in Aquatic Organisms**" 55th Midwest Theoretical Chemistry Conference (MWTCC55) | May 2025 | Wayne State University, Detroit, MI, USA
- "**PFAS Toxicities on Organisms Inhabiting the Great Lakes**" 54th Midwest Theoretical Chemistry Conference (MWTCC54) | May 2024 | University of Wisconsin-Madison, Madison, WI, USA
- "**Development of an Efficient Testing Framework for Interpreting Toxicity of Complex Mixtures for Ecological Risk Assessment**" SERDP Annual Meeting | August 2024 | Long Beach, CA, USA
- Amber Advanced Molecular Dynamics Workshop | August 2024 | San Diego Supercomputer Center (SDSC) at UC San Diego, La Jolla, CA, USA
- **"Effects of PFAS on Fish Proteins: A Computational Investigation"** American Chemical Society (ACS) CERM | June 2023 | Dearborn, MI, USA

PUBLICATIONS

- Bali, S.K.; Hall, K.; **Massoud, R.I**.; Almeida, N.M.S**. Interaction of Per- and Polyfluoroalkyl Substances with Estrogen Receptors in Rainbow Trout (Oncorhynchus mykiss): An *In Silico* Investigation.** Environmental Science & Technology, **2024**. DOI:10.1021/acs.est.4c03648
- **Massoud R.I.,** Bouaziz, M.; Abdallah, H.; Zeiz, A.; Flamini,G.; El-Dakdouki, M.H. **Comparative Study on the Chemical Composition and Biological Activities of the Essential Oils of Lavandula angustifolia and Lavandula x intermedia Cultivated in Lebanon.** ACS Omega, **2024**. DOI:10.1021/acsomega.4c00313
- **Massoud, R.I.,** et al**, Molecular Insights into PFAS-Induced Modulation of the Ecdysone Receptor in Aquatic Invertebrates.** Environmental Science & Technology, **2026**. *Submitted.*

EXTRACURRICULAR

- **Safety Officer** of the Wilson Group | August 2025 – July 2026
- **Head of MD Journal Club** of the Wilson Group | August 2024 – July 2025
- **Volunteer** at MSU Science Festival and recruitment events | 2023 – 2025
- **Volunteer Organizer** of the Midwest Quantum Collaboratory (MQC) | 2024
- **Member** of the Chemistry Graduate Student Organization at MSU | August 2022 – Present